# Geometry Review: Unit 5 — Right Triangles & Trigonometry

**Station 1 of 4:** The Pythagorean Theorem

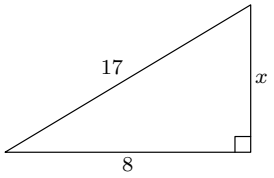
Name: _____ Date: _____

*Use $a^2 + b^2 = c^2$. Give exact answers in simplest radical form, and round decimals to the nearest tenth when asked. Diagrams are not drawn to scale.*

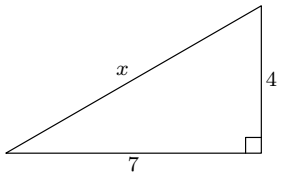
**1.** The legs of a right triangle are 6 and 8. Find the hypotenuse.

**2.** The legs of a right triangle are 5 and 12. Find the hypotenuse.

**3.** The hypotenuse of a right triangle is 17 and one leg is 8. Find the other leg.




**4.** The legs of a right triangle are 4 and 7. Find the hypotenuse. Give the exact answer and a decimal to the nearest tenth.

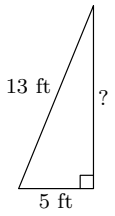



**5.** The hypotenuse of a right triangle is 9 and one leg is 5. Find the other leg in simplest radical form and as a decimal to the nearest tenth.

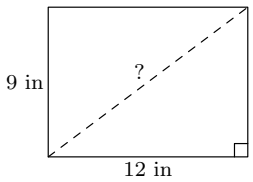
**6.** Is $9, 12, 15$ a Pythagorean triple? Explain.

**7.** A triangle has sides $7, 8, 12$. Is it acute, right, or obtuse?

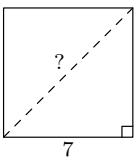
**8.** A 13-ft ladder leans against a wall with its base 5 ft from the wall. How high up the wall does it reach?




**9.** Find the length of the diagonal of a rectangle that is 9 in by 12 in.




**10.** Find the length of the diagonal of a square with side 7. Give the exact answer and a decimal to the nearest tenth.



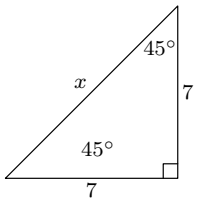
# Geometry Review: Unit 5 — Right Triangles & Trigonometry

**Station 2 of 4:** Special Right Triangles: 45°-45°-90° and 30°-60°-90°

Name: _____ Date: _____

*Give exact answers in simplest radical form (no decimals). Rationalize denominators. Diagrams are not drawn to scale.*

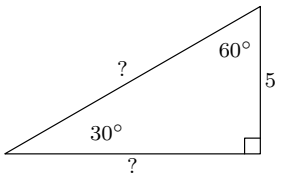
**1.** In a 45°-45°-90° triangle, each leg is 7. Find the hypotenuse.



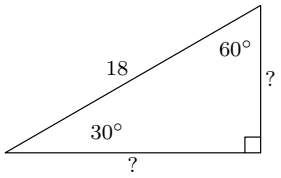

**2.** In a 45°-45°-90° triangle, the hypotenuse is 12. Find each leg.

**3.** In a 45°-45°-90° triangle, the hypotenuse is 10. Find each leg.

**4.** In a 30°-60°-90° triangle, the shorter leg is 5. Find the longer leg and the hypotenuse.




**5.** In a 30°-60°-90° triangle, the hypotenuse is 18. Find both legs.

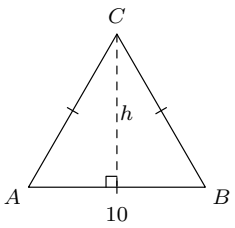



**6.** In a 30°-60°-90° triangle, the longer leg is $6\sqrt{3}$. Find the shorter leg and the hypotenuse.

**7.** In a 30°-60°-90° triangle, the longer leg is 9. Find the shorter leg and the hypotenuse.

**8.** The diagonal of a square is $8\sqrt{2}$. Find the side length.

**9.** Find the height of an equilateral triangle with side length 10.




**10.** Find the area of an equilateral triangle with side length 8.

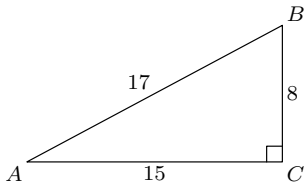
# Geometry Review: Unit 5 — Right Triangles & Trigonometry

**Station 3 of 4:** Trigonometric Ratios: Sine, Cosine, Tangent

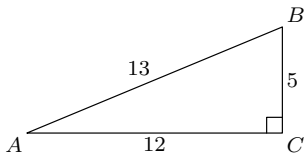
Name: _____ Date: _____

*Write ratios as simplified fractions. Remember SOH-CAH-TOA. Diagrams are not drawn to scale.*

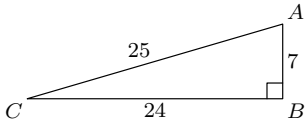
**1.** A right triangle has legs 8 and 15 and hypotenuse 17. $\angle A$ is opposite the leg of length 8. Find $\sin A$, $\cos A$, $\tan A$.




**2.** A right triangle has legs 5 and 12 and hypotenuse 13. $\angle B$ is opposite the leg of length 12. Find $\sin B$, $\cos B$, $\tan B$.




**3.** A right triangle has legs 7 and 24 and hypotenuse 25. $\angle C$ is opposite the leg of length 7. Find $\sin C$, $\cos C$, $\tan C$.




**4.** $\theta$ is acute and $\sin\theta = \dfrac{5}{13}$. Find $\cos\theta$ and $\tan\theta$.

**5.** $\theta$ is acute and $\cos\theta = \dfrac{3}{5}$. Find $\sin\theta$ and $\tan\theta$.

**6.** Complete: $\sin 20^\circ = \cos$ _____

**7.** In a right triangle, the side opposite $\theta$ is 6 and the side adjacent to $\theta$ is 8. Find $\tan\theta$, the hypotenuse, and $\sin\theta$.

**8.** Give the exact values: $\sin 30^\circ$, $\cos 60^\circ$, $\tan 45^\circ$, $\tan 60^\circ$.

**9.** In a right triangle, $\sin A = \dfrac{4}{5}$ and the side opposite $\angle A$ is 12. Find the hypotenuse and the adjacent side.

**10.** For the 3-4-5 triangle with $\angle A$ opposite 3, show that $\sin^2 A + \cos^2 A = 1$.

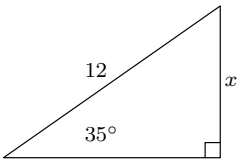
# Geometry Review: Unit 5 — Right Triangles & Trigonometry

**Station 4 of 4:** Solving Right Triangles with Trigonometry

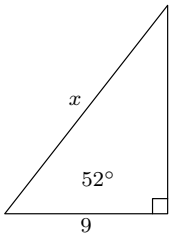
Name: _____ Date: _____

*Use a calculator in degree mode. Round lengths to the nearest tenth and angles to the nearest degree unless told otherwise. Diagrams are not drawn to scale.*

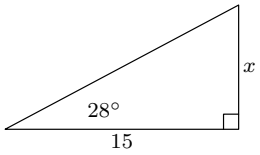
**1.** A right triangle has an angle of 35° and a hypotenuse of 12. Find the side $x$ opposite the 35° angle.



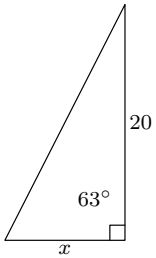

**2.** A right triangle has an angle of 52° and an adjacent side of 9. Find the hypotenuse $x$.




**3.** A right triangle has an angle of 28° and an adjacent side of 15. Find the opposite side $x$.




**4.** A right triangle has an angle of 63° and an opposite side of 20. Find the adjacent side $x$.

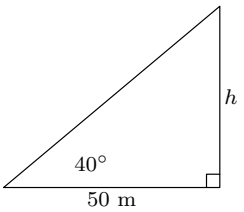



**5.** In a right triangle, the opposite side is 7 and the hypotenuse is 10. Find the angle $\theta$ to the nearest tenth of a degree.

**6.** In a right triangle, the adjacent side is 5 and the hypotenuse is 13. Find the angle $\theta$ to the nearest tenth of a degree.

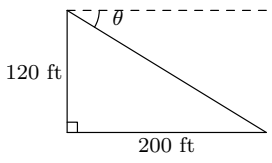
**7.** In a right triangle, the opposite side is 9 and the adjacent side is 4. Find the angle $\theta$ to the nearest tenth of a degree.

**8.** From a point 50 m from the base of a tower, the angle of elevation to the top is 40°. How tall is the tower?




**9.** A 20-ft ladder makes a 70° angle with the ground. How high up the wall does it reach?

**10.** From the top of a 120-ft cliff, a boat is 200 ft from the base of the cliff. Find the angle of depression to the boat, to the nearest tenth of a degree.



# Geometry Review: Unit 5 — Right Triangles & Trigonometry

Answer Key

### Station 1: The Pythagorean Theorem

1. 10
2. 13
3. 15
4. $\sqrt{65} \approx 8.1$
5. $2\sqrt{14} \approx 7.5$ $\quad \sqrt{81-25} = \sqrt{56}$
6. Yes; $9^2 + 12^2 = 81 + 144 = 225 = 15^2$
7. Obtuse $\quad 7^2 + 8^2 = 113 < 144 = 12^2$
8. 12 ft
9. 15 in
10. $7\sqrt{2} \approx 9.9$

### Station 2: Special Right Triangles: 45°-45°-90° and 30°-60°-90°

1. $7\sqrt{2}$
2. $6\sqrt{2}$ $\quad \dfrac{12}{\sqrt{2}} = \dfrac{12\sqrt{2}}{2}$
3. $5\sqrt{2}$
4. $5\sqrt{3};\ 10$
5. $9;\ 9\sqrt{3}$
6. $6;\ 12$
7. $3\sqrt{3};\ 6\sqrt{3}$ $\quad \dfrac{9}{\sqrt{3}} = 3\sqrt{3}$
8. 8
9. $5\sqrt{3}$
10. $16\sqrt{3}$ $\quad \frac{1}{2}(8)(4\sqrt{3})$

### Station 3: Trigonometric Ratios: Sine, Cosine, Tangent

1. $\dfrac{8}{17},\ \dfrac{15}{17},\ \dfrac{8}{15}$
2. $\dfrac{12}{13},\ \dfrac{5}{13},\ \dfrac{12}{5}$
3. $\dfrac{7}{25},\ \dfrac{24}{25},\ \dfrac{7}{24}$
4. $\cos\theta = \dfrac{12}{13},\ \tan\theta = \dfrac{5}{12}$
5. $\sin\theta = \dfrac{4}{5},\ \tan\theta = \dfrac{4}{3}$
6. $70°$ $\quad$ *sine and cosine of complementary angles are equal*
7. $\tan\theta = \dfrac{3}{4};\ \text{hyp} = 10;\ \sin\theta = \dfrac{3}{5}$
8. $\dfrac{1}{2},\ \dfrac{1}{2},\ 1,\ \sqrt{3}$
9. $15;\ 9$
10. $\left(\dfrac{3}{5}\right)^2 + \left(\dfrac{4}{5}\right)^2 = \dfrac{9}{25} + \dfrac{16}{25} = 1$

### Station 4: Solving Right Triangles with Trigonometry

1. $x = 12\sin 35° \approx 6.9$
2. $x = \dfrac{9}{\cos 52°} \approx 14.6$
3. $x = 15\tan 28° \approx 8.0$
4. $x = \dfrac{20}{\tan 63°} \approx 10.2$
5. $\theta = \sin^{-1}\left(\dfrac{7}{10}\right) \approx 44.4°$
6. $\theta = \cos^{-1}\left(\dfrac{5}{13}\right) \approx 67.4°$
7. $\theta = \tan^{-1}\left(\dfrac{9}{4}\right) \approx 66.0°$
8. $50\tan 40° \approx 42.0$ m
9. $20\sin 70° \approx 18.8$ ft
10. $\tan^{-1}\left(\dfrac{120}{200}\right) \approx 31.0°$ $\quad$ *angle of depression = angle of elevation from the boat*